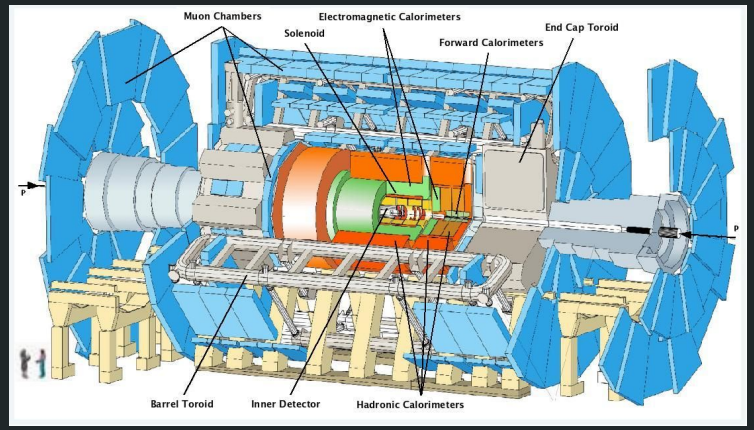
Equivariant Edge Classification for Particle Track Reconstruction

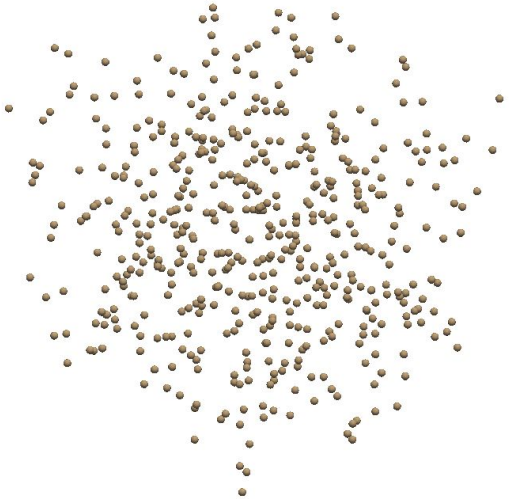
Russell Castro for BeBOP 5/7/2025

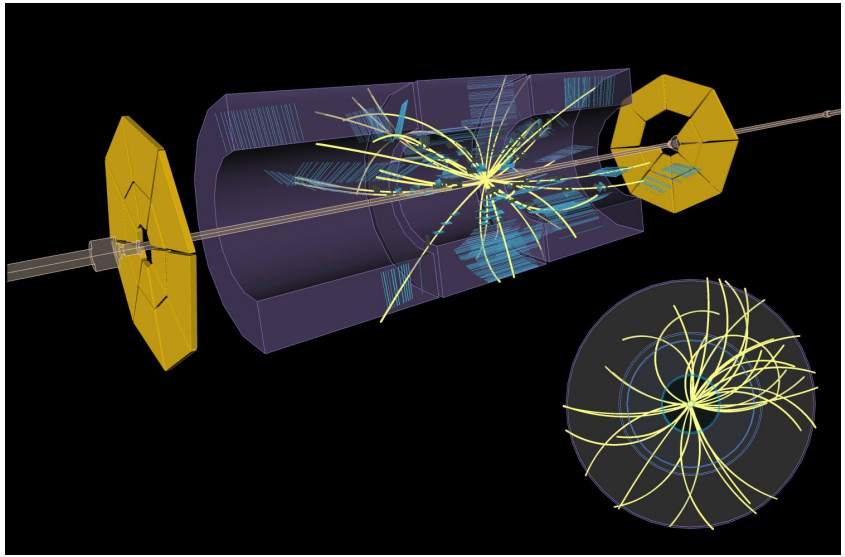
Problem Space



Reconstruction Experimental Chain

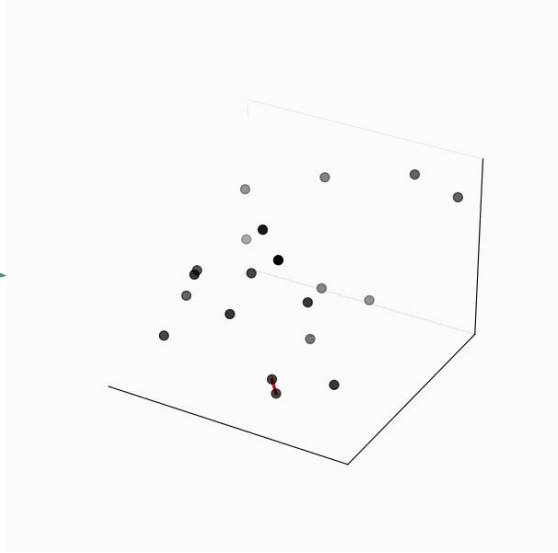
raw hit cloud $O(10^5 - 10^6)$ → edge classification → track candidate construction → particle identification





Prior Work by Ju et. al. (2021)

1. Pre-processing
2. Embedding and Fixed-radius NN (trainable)
3. Edge filter (trainable)
4. **Edge Classification: InteractionGNN (trainable)**
5. Connected components (true/fake classification)
6. Post-processing



Motivation

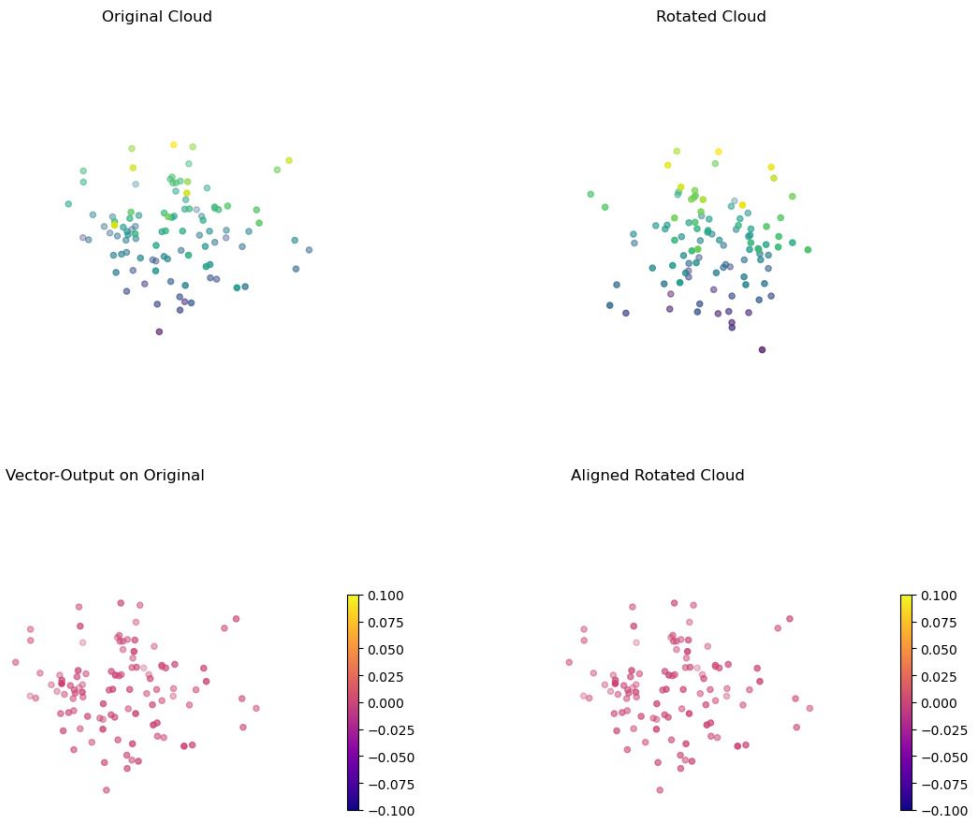
1. Track-building dominates High-Energy Physics (HEP) reconstruction runtime.
2. Is there a more efficient way of capturing geometry?
3. Lower parameter count?

Equivariance

Equivariance

$$\mathcal{L} \circ D^{\mathcal{X}}(g) = D^{\mathcal{Y}}(g) \circ \mathcal{L}$$

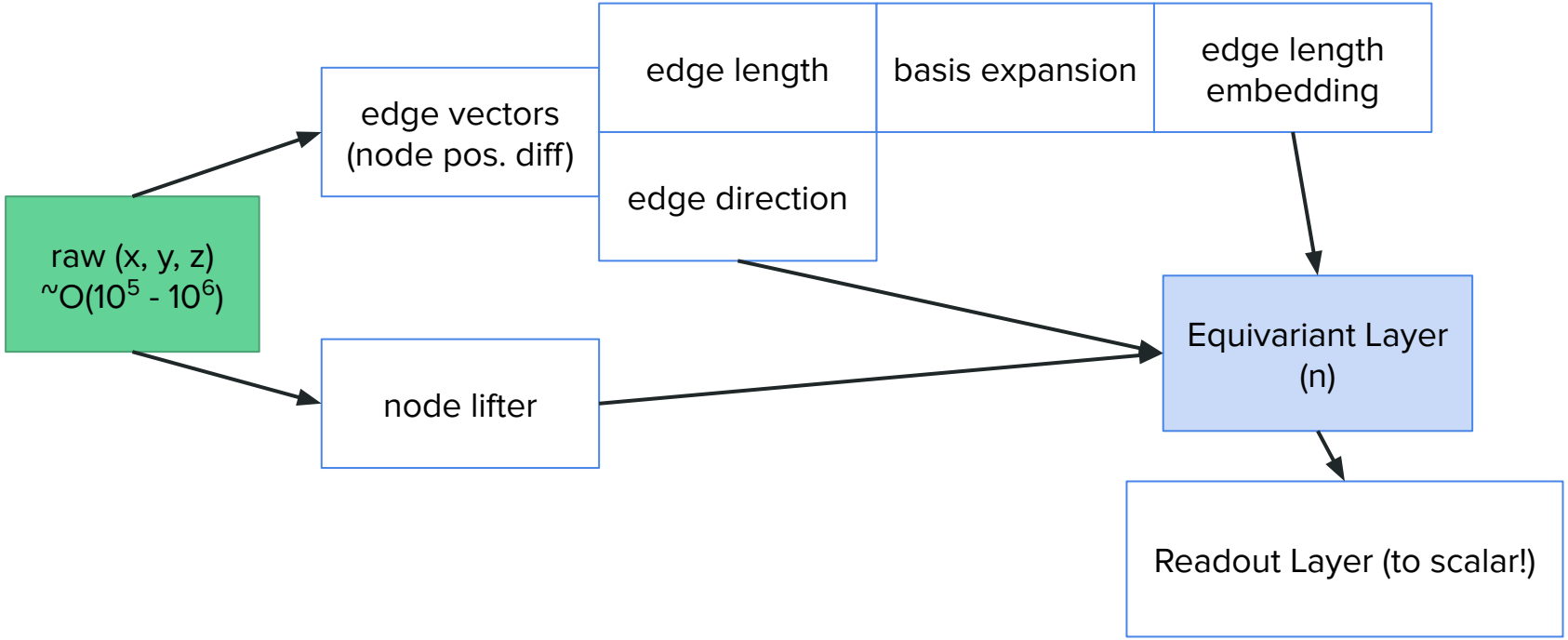
- An equivariant function respects physical symmetries (i.e. rotations, reflections)
- Condensed structure reduces complexity
- Consistent outputs under coordinate transforms


Original Cloud
Rotated Cloud
Vector-Output on Original
Aligned Rotated Cloud
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100

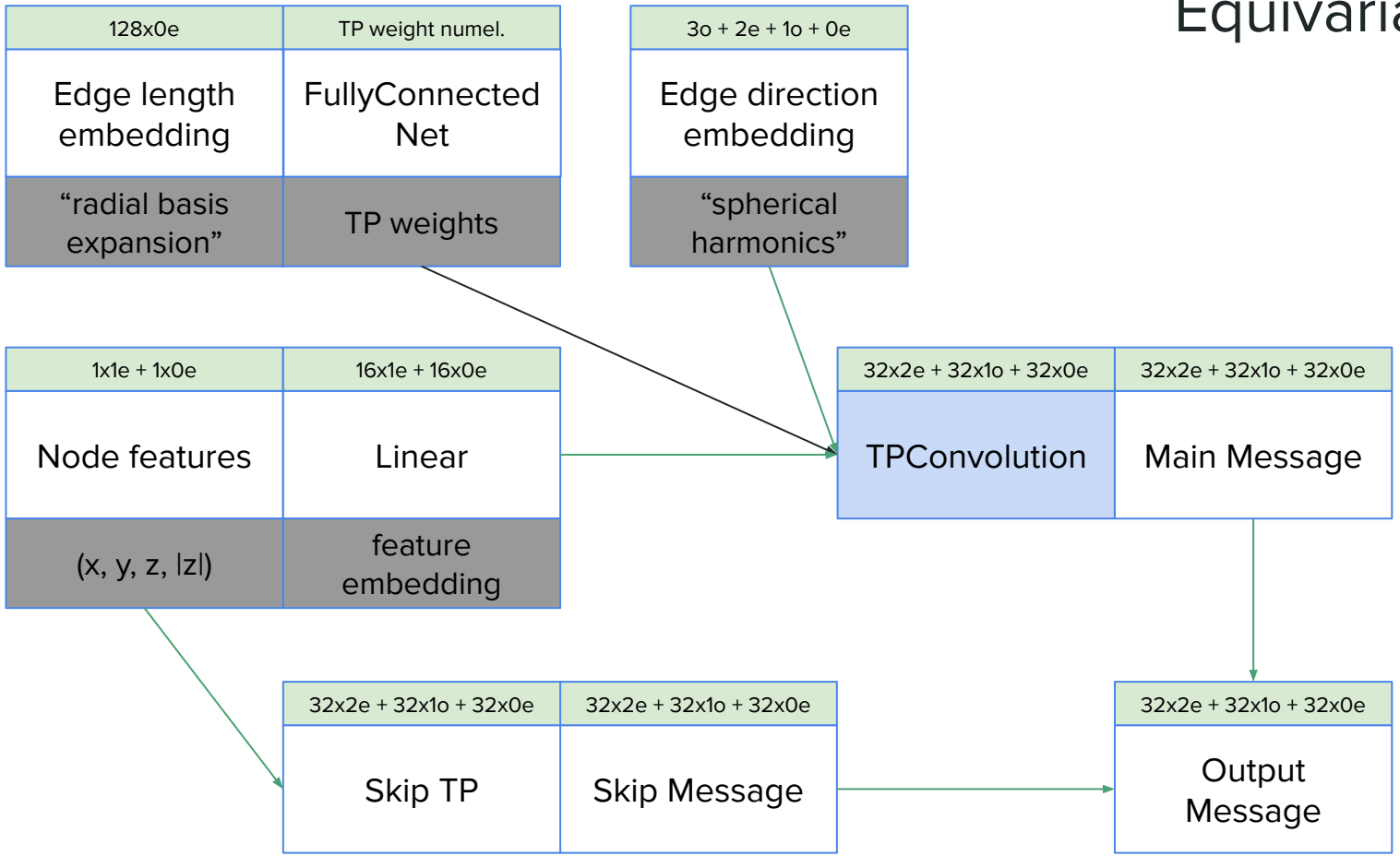
Why Equivariance Here?

- Detector hits are observed in arbitrary orientations.
- Spherical-harmonics filters with Clebsch-Gordan algebra → exact O(3) equivariance.
  - Combined features are fused with CG coefficients to enforce coupling rules (i.e. triangle inequalities, parity, etc.)
- Remove costly angular augmentation; parameter-efficient learning.

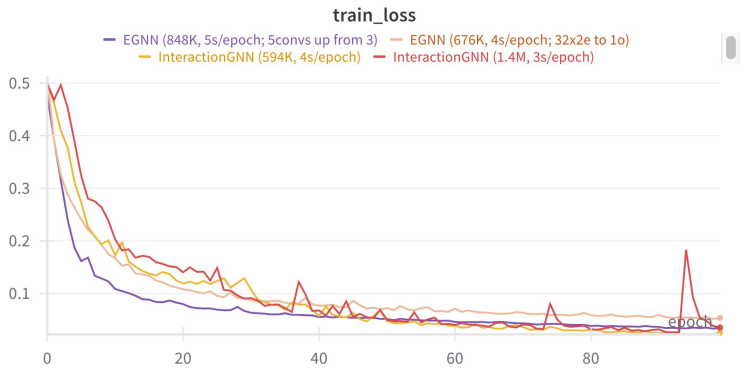
EGNN Architecture

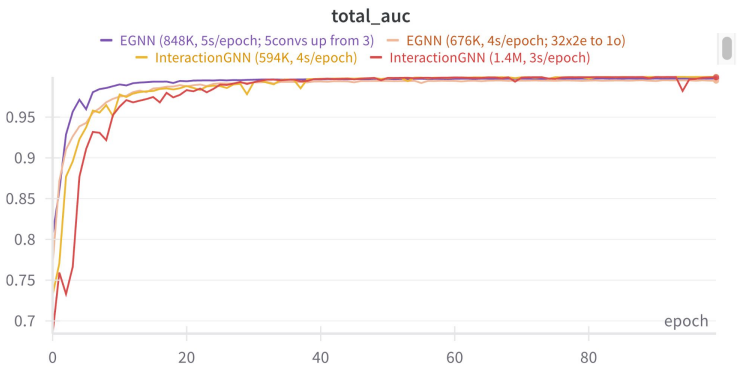


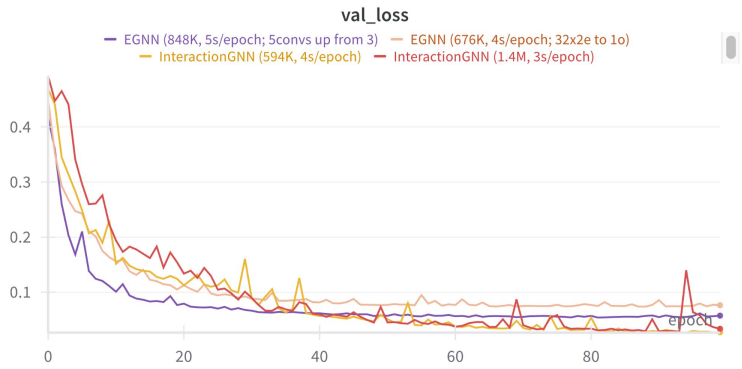
Equivariant Layer

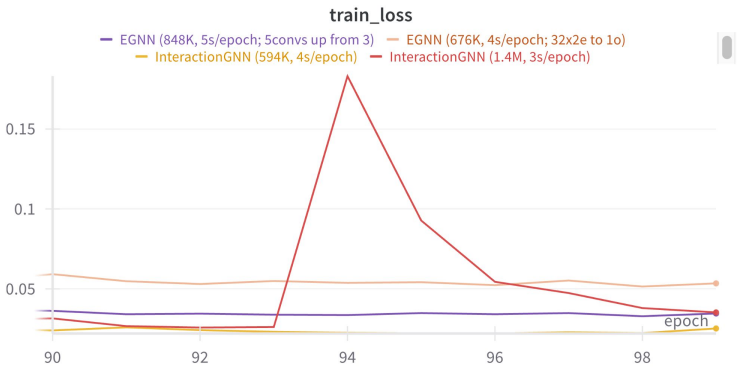


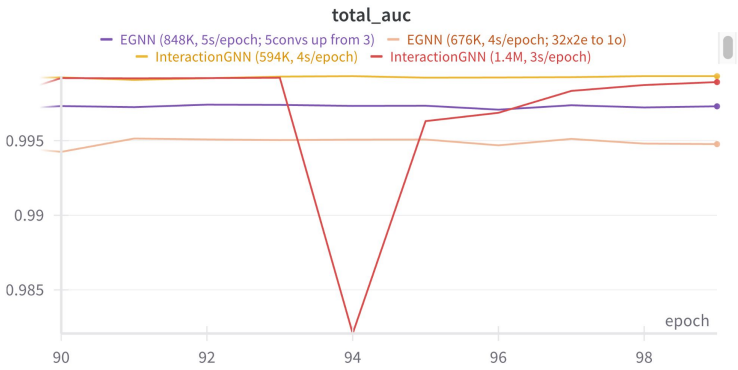
Results Against InteractionGNN Baseline

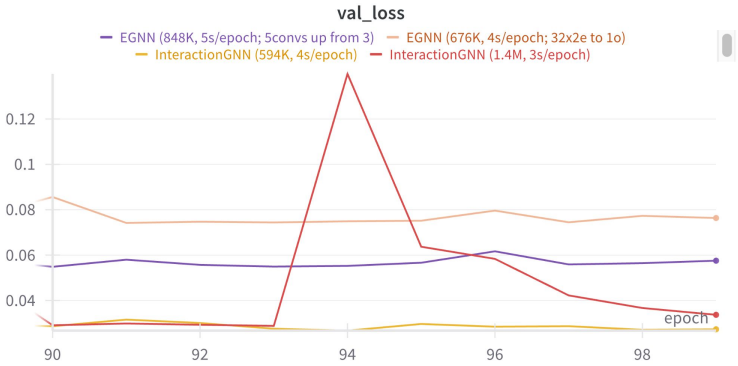




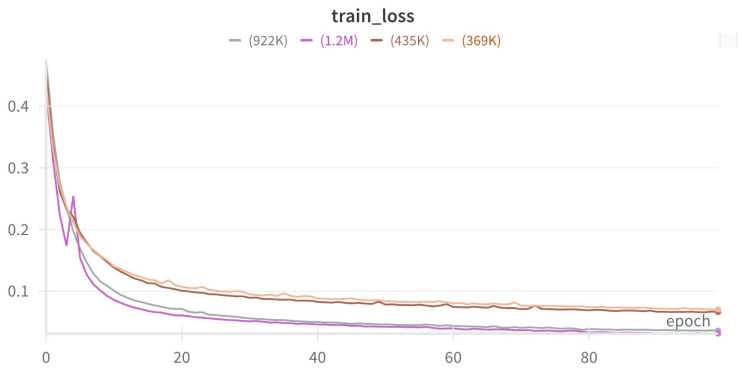


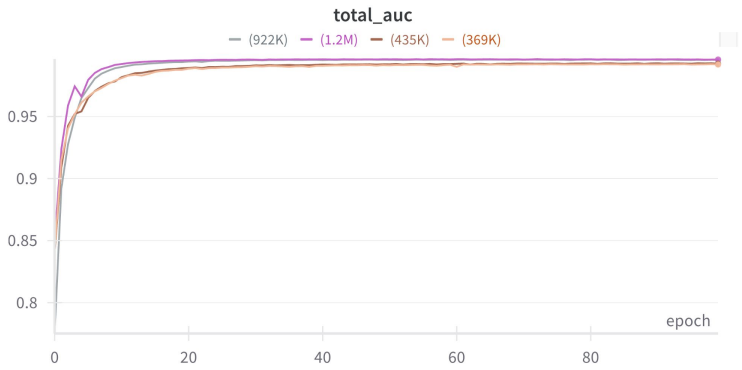

train_loss
EGNN (848K, 5s/epoch; 5convs up from 3)
EGNN (676K, 4s/epoch; 32x2e to 1o)
InteractionGNN (594K, 4s/epoch)
InteractionGNN (1.4M, 3s/epoch)
0.15
0.1
0.05
epoch
90
92
94
96
98

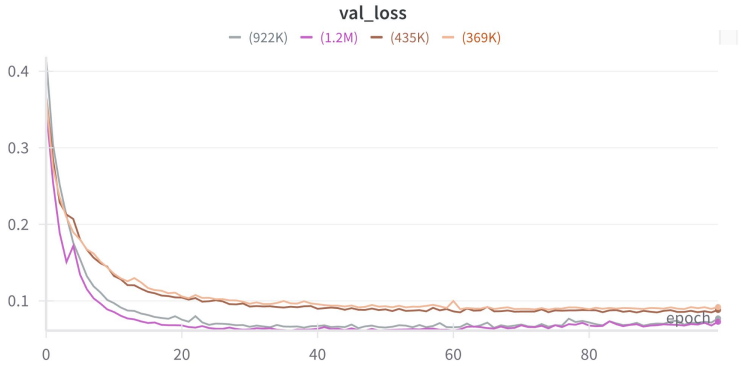

total_auc
EGNN (848K, 5s/epoch; 5convs up from 3)
EGNN (676K, 4s/epoch; 32x2e to 1o)
InteractionGNN (594K, 4s/epoch)
InteractionGNN (1.4M, 3s/epoch)
0.995
0.99
0.985
epoch
90
92
94
96
98

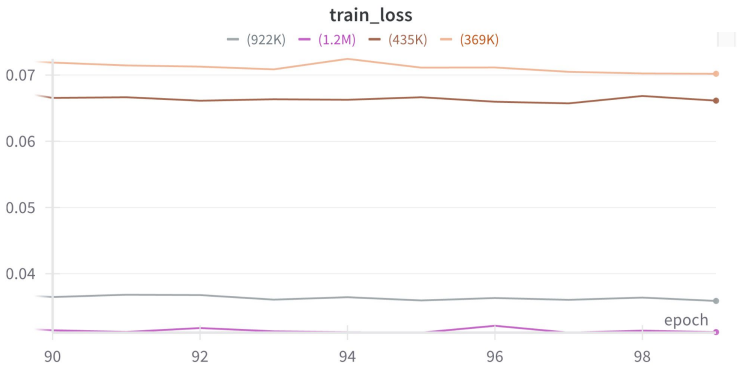

val_loss
EGNN (848K, 5s/epoch; 5convs up from 3)
EGNN (676K, 4s/epoch; 32x2e to 1o)
InteractionGNN (594K, 4s/epoch)
InteractionGNN (1.4M, 3s/epoch)
0.12
0.1
0.08
0.06
0.04
epoch
90
92
94
96
98

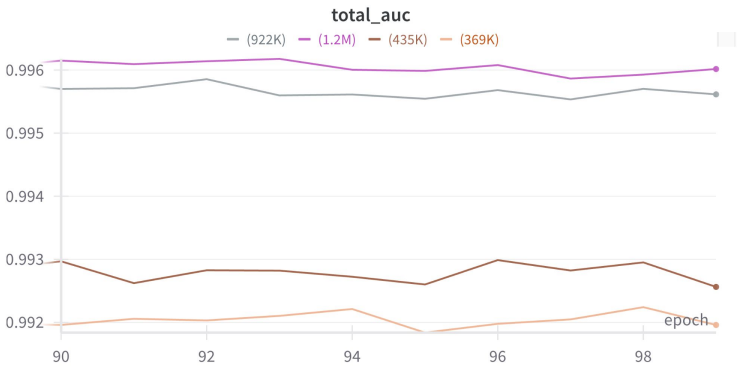
Scaling with Parameters

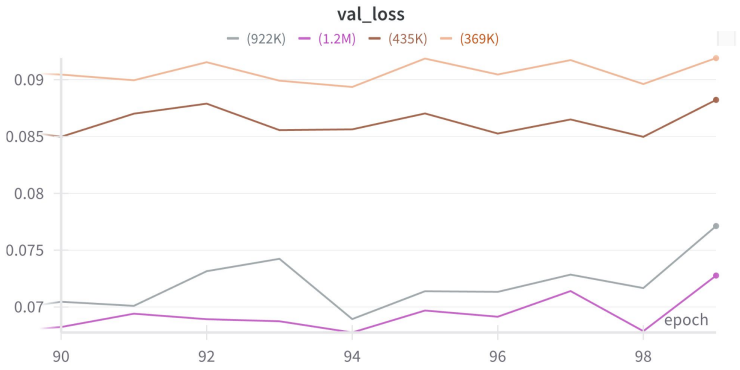







train_loss
(922K)
(1.2M)
(435K)
(369K)
0.07
0.06
0.05
0.04
epoch
90
92
94
96
98


total_auc
(922K)
(1.2M)
(435K)
(369K)
0.996
0.995
0.994
0.993
0.992
epoch
90
92
94
96
98


val_loss
(922K)
(1.2M)
(435K)
(369K)
0.09
0.085
0.08
0.075
0.07
epoch
90
92
94
96
98

Thank you!

$$W(\mathbf{r}) = \sum_{\ell,m} R_\ell(\|r\|)\, Y_\ell^m(\hat{\mathbf{r}})$$

$$W(x) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} R_\ell(\|x\|)\, Y_\ell^m(\hat{x})\, C_{\ell m}$$